Monitoring Report

MedcoEnergi Associated Gas Recovery and Utilization Project

Monitoring period – 28/03/2007 to 27/03/2008 (both days included)

Monitoring report No.2
Date and version no. 22/06/2010, version 2.1

TABLE OF CONTENT

1. Project Background

The Project aims to reduce CO_2 emission by capturing and processing associated gas that would otherwise have been flared. Associated gas is produced as a by-product of oil production activity at Kaji-Semoga oil field, operated by PT Medco E&P Indonesia (MEPI), located ± 80 km from Palembang, the capital city of South Sumatera province, Indonesia, and flared on-site. The captured gas is processed in an LPG plant. The installation of the LPG plant allows the recovery and processing of the associated gas into LPG, condensate, and residue gas.

The host country is Indonesia and the project participant is PT. Medco LPG Kaji. The LPG plant is located at the latitude and longitude of 02° 49’ 30” S, 104 º 04’ 30” E, approximately 30 km from Sekayu, the capital city of Musi Banyuasin Regency and 80 km west of Palembang, the capital city of South Sumatera Province.

2. Project Implementation and sustainable development

2.1 Implementation status

The Project has been completed and commercialized since March 2004. The LPG plant has been running generating a total of 140,046 carbon credit for the second monitoring period from 28 March 2007 to 27 March 2008.
The emission reduction credits are calculated and reported in more details in Section 4 of this monitoring report. These credits are to be verified and claimed as Voluntary Carbon Units (VCUs).

2.2 Operation of the project

Overall the power plant has been successfully managed and operated by a team of competent personnel. Its performance has been satisfactorily and in line with the design. The design has not changed compared to the PD.

2.3 Sustainable development

The project has strongly contributed on the sustainable development at social, economic, environmental and technological aspects as explained below:

Social well-being:
- The project contributes to the development of the region by increasing community development and corporate social responsibility of PT. Medco LPG Kaji.
- During the construction and operation, there are various kinds of work, which generate employment on regular and permanent basis.

Economic well-being:
- The project activity generates employment in the local area.
- The project activity also leads to diversification of the national energy supply, where such waste gas could be recovered and used as an energy source to replace conventional fossil fuel.

Environmental well-being:
- The project reuses the associated gas from oil exploration for LPG production, which otherwise would have flared. The project activity is meant to reduce methane and CO_2 emissions (GHG emissions), which are traditionally generated by flaring the associated gas.

- The gas recovery unit produces dry gas, LPG and condensate, which will be used to replace conventional-mined fossil fuel. This is increasingly important given the higher demand for LPG in the country, associated with the government fuel-switching program (from kerosene to LPG).[1]
- Thus, the project causes no negative impact on the surrounding environment; and in the end contributes to environmental well-being.

Technological well-being:
- The project supports technology transfer, as well as contributes to capacity building of the labor force through trainings and practical works.
- The project utilizes local products developed in the region, when replacement of spare parts is necessary.

2.4 Actual emission reductions

The second monitoring period is from 28 March 2007 to 27 March 2008. During the period, the actual emission reduction is 140,046 ton CO_2. Detailed calculations of the emissions can be found in section 4 of this monitoring report.

3. Compliance with the monitoring methodology

The project personnel and management has conducted the monitoring activities by following the monitoring plan presented in the PD and in accordance with the following approved methodology and tool:

Applied methodology

AM0009 – Recovery and utilization of gas from oil wells that would otherwise be flared or vented (Version 04, 25th March 2009, EB 46).

Applied tool

"Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion" (version 02)

3.1 Monitoring period

The second monitoring period covers 28 March 2007 to 27 March 2008 (both days included)

3.2 Monitoring parameters

The parameters monitored applied for the determination of the emission reductions is described in detail in section 3.3 of the Project Document v.2 dated 5th September 2009.
The simplified diagram of all measurement devices installed and their relative position is presented in Figure 1 below. Primary instruments are shown with numbers. The one with no number (or NN) will be installed and will start to operate latest in April 2010. The primary instruments includes:

- Total feed gas supplied to MLK, as measured by gas flow meter FTP-1011. Other supporting feed gas orifice meters are FE-1011, FQR-1011 and FIQ-1011.
- Total dry gas produced the sum of measurement by gas flow meter FTP-2014 and FTP-2852.
- Total dry gas delivered back to MEPI (also called 'lean gas' in MEPI-MLK monthly report and called 'residue gas' in the laboratory test sheet), as measured by gas flow meter FTP-2014.
- Total dry gas used as fuel for own consumption (also called 'fuel gas'), as measured by gas flow meter FTP-2852.

[1] Presidential Decree no. 104 year 2007 on the Logistics, Distribution, and Price On Liquefied Petroleum Gas.

- Total LPG produced, as measured by flow meter FT-2122. LPG is stored in four LPG storage tanks, and a level transmitter indicates the level height for each tank.
- Total condensate produced, as measured by flow meter FT-2151.

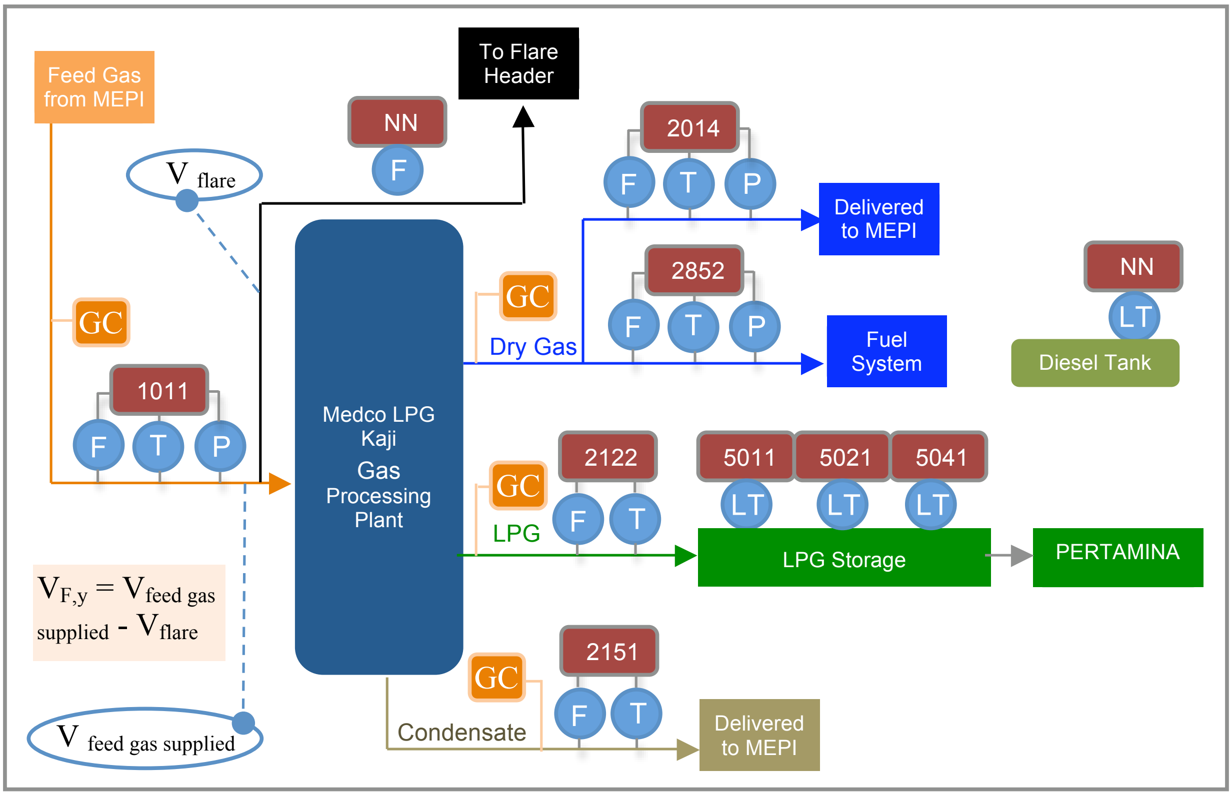

Figure 1 Project's Simplified Schematic Diagram, including relative position of its metering devices

Error! Reference source not found. below summarizes parameters that are monitored against requirement in the PDD. Readings from all instruments listed in this table are logged on the basis as mandated by the PD.

Table 1. Key monitored parameters, its relevant instrument, and estimated values in the PD.

PD Ref	Output of measurements	Unit	Source of Data as required in section 3.3 of PD	Instruments Installed / Instrument's Accuracy	Comparative medium for cross checking
$V_{F,y}$	Monthly amount of total recovered gas before the recovered gas is delivered to gas processing plant	Nm^3	Calculated from on-site measurement on the recovered gas supplied meter and on the gas-flaring stream before the recovered gas is delivered to gas processing plant	N/A. No instrument installed.	N/A
$NCV_{RG,F,y}$	Monthly average net calorific value of recovered gas	TJ/Nm^3	Calculated weighted average net calorific value using measured recovered gas composition (called 'feed gas' in the lab test result on gas composition)	Gas Chromatography 95%	N/A
$FC_{dry\ gas,y}$	Monthly amount of dry gas combusted for power generation for internal consumption	m^3/y	Onsite measurements using flow meter	*FTP-2852* Gas meter measuring the dry gas used for power generation for internal consumption F – 2852: 0.1 in H_2O T – 2852: 0.001 P – 2852: 0.025% FS	N/A
$NCV_{dry\ gas,y}$	Monthly average net calorific value of dry gas	GJ/m^3	Calculated weighted average net calorific value using measured dry gas composition (called 'residue gas' in the lab test result on gas composition)	Gas Chromatography 95%	N/A

<table>
<tr><td>$EF_{CO2,dry\ gas,y}$</td><td>Weighted average CO_2 emission factor of dry gas</td><td>tCO_2/GJ</td><td>2006 IPCC Guidelines for National Greenhouse Gas Inventories: Workbook volume 2 chapter 1 Table 1.4: IPCC default value: 56’100</td><td>N/A. No instrument installed.</td><td>N/A</td></tr>
<tr><td>$FC_{diesel\ fuel,y}$</td><td>Quantity of diesel fuel combusted in diesel gensets</td><td>litre/y</td><td>Onsite measurements using volume meters[2]
Purchase orders that were used in the diesel fuel combusted calculation are following:
<table>
<tr><th>PO No.</th><th>PO Date</th><th>Quantity (litre)</th></tr>
<tr><td>4400002066</td><td>4 Apr 07</td><td>5’000</td></tr>
<tr><td>4400002067</td><td>20 Apr 07</td><td>5’000</td></tr>
<tr><td>4400002068</td><td>4 May 07</td><td>5’000</td></tr>
<tr><td>4400002069</td><td>19 May 07</td><td>5’000</td></tr>
<tr><td>4400002070</td><td>5 June 07</td><td>5’000</td></tr>
<tr><td>4400002071</td><td>20 June 07</td><td>5’000</td></tr>
<tr><td>4400002072</td><td>5 Jul 07</td><td>5’000</td></tr>
<tr><td>4400002073</td><td>7 Aug 07</td><td>5’000</td></tr>
<tr><td>4400002074</td><td>10 Nov 07</td><td>5’000</td></tr>
<tr><td>4400002075</td><td>6 Dec 07</td><td>5’000</td></tr>
<tr><td>4400002077</td><td>29 Jan 08</td><td>5’000</td></tr>
<tr><td>4400002217</td><td>5 Mar 08</td><td>5’000</td></tr>
<tr><td>4400002218</td><td>15 Mar 08</td><td>5’000</td></tr>
</table></td><td>Using level indicator
Meter will be installed and ready to operate in April 2010.</td><td>Purchase orders from the purchasing departments.</td></tr>
<tr><td>$\rho_{diesel\ fuel,y}$</td><td>Density of diesel fuel</td><td>ton/litre</td><td>Diesel fuel supplier: PT. Pertamina (Persero): 0.000815</td><td>N/A. No instrument installed.</td><td>N/A</td></tr>
<tr><td>$NCV_{diesel\ fuel,y}$</td><td>Weighted average net calorific value of diesel fuel</td><td>TJ/ton</td><td>2006 IPCC Guidelines for National Greenhouse Gas Inventories: Workbook volume 2 chapter 1 Table 1.2: IPCC default value: 0.043</td><td>N/A. No instrument installed.</td><td>N/A</td></tr>
<tr><td>$EF_{CO2,diesel\ fuel,y}$</td><td>Weighted average CO_2 emission factor of diesel fuel</td><td>tCO_2/GJ</td><td>2006 IPCC Guidelines for National Greenhouse Gas Inventories: Workbook volume 2 chapter 1 Table 1.4: IPCC default value: 74’100</td><td>N/A. No instrument installed.</td><td>N/A</td></tr>
<tr><td>$V_{flare,\ y}$</td><td>Monthly volume of gas flared at flaring stream</td><td>Nm^3</td><td>Onsite measurements using flow meter. Until the flow meter on the flaring stream is installed, the flared gas stream is calculated according to equations in section 4.2 VCS PD.</td><td>Digital flow meter for flare gas (model: GF868 from GE)
Metering device is ordered and installation preparation is done. Meter is ready to operate in April 2010.</td><td>Volume balance on the gas inflow and the gas outflow</td></tr>
<tr><td>$V_{feed\ gas\ supplied,\ y}$</td><td>Monthly amount of feed gas supplied to the gas processing plant</td><td>m^3</td><td>Onsite measurements using flow meter</td><td>FTP-1011
Gas meter measuring the feed gas delivered into the plant
F – 1011: 0.025 % FS
T – 1011: 0.025 % FS
P – 1011: 0.025 % FS</td><td>Feed Gas Monthly Report signed by MEPI and MLK</td></tr>
<tr><td>$V_{dry\ gas,\ y}$</td><td>Monthly amount of dry gas produced in the gas processing plant</td><td>m^3</td><td>Onsite measurements using flow meter</td><td>FTP-2014
Gas meter measuring the dry gas delivered directly to MEPI (also called ‘lean gas’)
FTP-2852
Gas meter measuring the dry gas used for power generation for internal consumption (also called ‘fuel gas’)
Total dry gas produced is the sum of both dry gas streams.</td><td>N/A</td></tr>
</table>

[2] As per validated VCS PD, the quantity of diesel fuel combusted should be measured onsite using volume meters. However, between 28 March 2007 until 27 March 2008 no such measuring instrument has been installed as the meter will only be installed and operated starting April 2010. As an alternative calculation approach purchase orders of diesel during the verification period have been collected and conservatively applied as total diesel usage of the installed gensets.

				FTP 2852 please see $FC_{dry\ gas,y}$ F – 2014: 0.1 in H_2O T – 2014: 0.01 P – 2014: 0.01 psi	
$V_{LPG,\ y}$	Monthly amount of LPG produced in the gas processing plant	ton	Onsite measurements using metering device	*FT– 2122* Turbine meter measuring the LPG produced. *LT-5011/5021/5041* Tank level transmitter measuring LPG stored in the storage F – 2122: 0.1 T – 2122: 0.01	N/A
$\rho_{LPG,y}$	Average density of LPG	ton/m^3	Onsite measurements using Gas Chromatography	Gas Chromatography 95%	N/A
$V_{condensate,\ y}$	Monthly amount of condensate produced in the gas processing plant	barrel	Onsite measurements using flow meter	*FT– 2151* Flow meter measuring the condensate produced F – 2151: 0.1 T – 2151: 0.01	N/A

3.3 Monitoring management and operational system

The monitoring activities have been conducted by qualified and experienced personnel in accordance with the monitoring plan presented in the PD.

The technical manager (plant manager) is in charge of the overall monitoring functions to monitor key and related parameters in the calculation of the GHG emission reductions by the project activity.
Throughout the line-of-command from operators to supervisors, to site monitoring office and to the technical manager, the accuracy of data collections and management is ensured by internal audits.
DCS (Digital Controlling System) also constantly on a real-time basis or verifies the amount of feed gas supplied to and all products (dry gas, LPG and condensate) from the gas processing plant.
Details about the monitoring management including roles and responsibilities of related personnel, QA/QC of parameters monitored and procedures for accounting for missing key parameters and other details can be found in the SOP Monitoring of Emission Reduction MLK-PLT-PROD-001.

3.4 Accuracy of monitoring equipments

The accuracy of the monitoring results are in conformity with calibration requirements, recording frequency and quality assurance and quality control procedure stated in the monitoring plan.

3.4.1 Calibration of monitoring equipments

The following is the calibration status of the monitoring equipment during this period:

Meter Tag Number	Parameter measured	Duration of calibration mentioned in the PD	Latest calibration date	By	Next calibration date
F 1011 (Flow transmitter) *T 1011 (Temperature transmitter)* *P 1011 (Pressure transmitter)* *FE 1011 (Orifice plate)* *FQR 1011 (Three pen recorder)* *FIQ 1011 (Flow computer)*	Feed gas supplied to gas processing plant	Maximum every 2 years	14 March 2007 18 March 2008	Direktorat Jenderal Minyak dan Gas Bumi (General Directorate of Oil and Gas) Certificate number: 316/SMG/18.06/DMT/2007 92/30/SMG/18.06/DJM/2008	18 March 2009
Gas Chromatography	Feed gas & dry gas composition, density LPG	Maximum every 2 years	31 March 2007 27 February 2008	Laboratorium Kalibrasi LEMIGAS (Oil and Gas Technology Research & Development Centre) Certificate number:	27 February 2009

				SV-1215 GC Manufacturer: PT. Berca Niaga Medika Certificate number: n/a	
F 2014 (Flow transmitter) *T 2014(Temperature transmitter)* *P 2014 (Pressure transmitter*	Dry Gas produced delivered to MEPI	Maximum every 2 years	11 December 2007	PT. Morita Tjokro Gearindo Certificate number: F 2014: P.351.1207 T 2014: T.435.1207 P 2014: P.352.1207	28 April 2009
F 2852 (Flow transmitter) *T 2852 (Temperature transmitter)* *P 2852 (Pressure transmitter*	Total fuel (dry gas) combusted for own consumption	Maximum every 2 years	11 December 2007	PT. Morita Tjokro Gearindo Certificate number for: F 2852: P.349.1207 T 2852: T.434.1207 P 2852: P.350.1207	29 April 2009
F 2122 (Flow transmitter) *T 2122 (Temperature transmitter)*	LPG produced from the LPG plant	Maximum every 2 years	12 December 2007	PT. Morita Tjokro Gearindo Certificate number for: F 2122: E.601.1207 T 2122: T.437.1207	29 April 2009
F 2151 (Flow transmitter) *T 2151 (Temperature transmitter)*	Condensate produced from the LPG plant	Maximum every 2 years	12 December 2007	PT. Morita Tjokro Gearindo Certificate number for: F 2151: E.602.1207 T 2151: T.438.1207	29 April 2009
LT 5011 (Level transmitter) *LT 5021 (Level transmitter)* *LT 5031 (Level transmitter)* *LT 5041 (Level transmitter)*	LPG stored in LPG storage	Maximum every 2 years	6 – 8 April 2007	Internal Calibration by Medco	Internal calibration: 5 March 2009 External calibration: 1 May 2009

3.4.2 Monitoring equipment malfunctions

There was no metering devices breakdown or malfunction. All the meters listed above have performed well. However, any plant shut down or any other malfunctions such as DCS display off are recorded in the logbook. Provisions to handling the missing parameters are clearly stated in the monitoring procedures should there be any parameter loss.

4. Emission Reduction calculations

The formulas used for calculation are in accordance with the approved methodology AM0009 – Recovery and utilization of gas from oil wells that would otherwise be flared or vented (Version 04, 25th March 2009, EB 46). The monitoring report covers the period of 28 March 2007 to 27 March 2008 (both days included).

4.1 Baseline Emissions

4.1.1 Formula used for baseline emissions calculation

The formula used for determination of the baseline emissions are described in section 4.3 of the Project Document v.2 dated 5th September 2009.

Baseline emissions are calculated as follows:

Parameter	Description	Unit	Value	Source
BE_y	Baseline emission during the period y	tCO_2e	Calculated	Equation (1)
$V_{F,y}$	Volume of total recovered gas measured used on-site, during period y	Nm^3	Measured	MLK
$NCV_{RG,F,y}$	Net calorific value of recovered gas measured at point F in Figure 1 during the period y	TJ/ Nm^3	Measured	MLK
$EF_{CO2,methane}$	CO_2 emission factor for methane	tCO_2/TJ	49.55	AM0009/Version04

$$BE_y = V_{F,y} \times NCV_{RG,F,y} \times EF_{CO2,methane} \qquad (1)$$

Baseline emissions calculated from gas projection refer to equation (1).

Figure 3 below shows the conditions at the gas processing plant in the feed gas area.

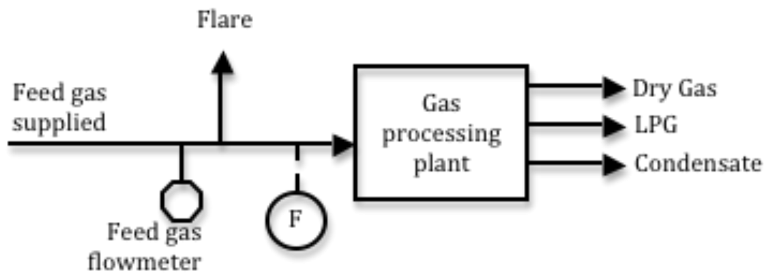

Figure 3. Recovered gas stream to the gas processing plant

As shown above, the flare stream (for safety reason) is put after the recovered gas flow meter before delivered the main equipments. Therefore, the feed gas rate into the gas processing plant at point F, $V_{F,y}$, is calculated as follow:

$V_{F,y} = V_{feed\ gas\ supplied} - V_{flare}$ Feed gas equation 1)

$V_{F,y} = V_{dry\ gas,y} + V_{LPG,y} + V_{condensate,y}$ Feed gas equation 2)

For calculation, Feed gas equation 2) is used.

4.1.2 Baseline emissions calculation

Data input

Date		Month	Year	$V_{feed\ gas\ supplied,y}$ (mmscf)	$V_{dry\ gas,y}$*		$V_{LPG,y}$	$V_{condensate,y}$
From	To				Delivered back to MEPI (mmscf)*	Total Fuel-own cons. (mmscf)**	ton	barrel
28	31	Mar	2007	57.84	32.93	8.80	356.24	1,547.84
1	30	Apr	2007	378.09	206.63	63.09	2,438.04	9,081.78
1	31	May	2007	335.87	176.45	65.03	2,195.80	7,855.16
1	30	Jun	2007	330.30	168.73	61.57	2,125.93	7,456.31
1	31	Jul	2007	324.26	168.30	63.96	2,086.66	7,094.43
1	31	Aug	2007	321.69	159.90	63.57	2,106.79	7,171.81
1	31	Sep	2007	312.38	156.43	61.58	2,025.75	6,852.67
1	30	Oct	2007	306.60	150.50	63.60	1,974.29	6,812.41
1	31	Nov	2007	307.97	166.41	59.43	1,746.35	6,323.26
1	30	Dec	2007	300.14	162.17	62.77	1,730.53	5,661.13
1	31	Jan	2008	289.06	141.57	66.73	1,657.98	5,897.87
1	31	Feb	2008	214.51	119.92	52.62	1,300.31	4,812.74
1	27	Mar	2008	188.28	106.64	49.39	1,127.06	4,399.09
		2007	**TOTAL**	2,975.14	1,548.45	573.40	18,786.38	65,856.80
		m^3	**TOTAL**	84,244,065	43,845,911	16,236,395	34,938	10,471
		2008	**TOTAL**	691.85	368.13	168.74	4,085.35	15,109.70
		m^3	**TOTAL**	19,590,510	10,423,970	4,778,042	7,598	2,403

* $V_{dry\ gas}$ is the total sum of dry gas delivered to MEPI and total fuel for internal consumption.
* Dry gas delivered to MEPI is also called lean gas.
* ** Total fuel is the dry gas produced from the LPG plant that is directly used for internal consumption at the Power House (= $FC_{dry\ gas}$). Dry gas used as fuel for own consumption is also called fuel gas.

Thus, the gas flow balance between input and output is as follows:

Parameter		Unit	Value	Source
Input: Feed Gas Supply	$V_{feed\ gas\ supplied,y}$	m^3	103,834,575	Measurement
Output: Total Product*	$V_{F,y}$	m^3	75,339,728	Feed gas equation 1)
Output: Flare	$V_{flare,y}$	m^3	28,494,847	Feed gas equation 2)

* Total product is the sum of dry gas, LPG and condensate produced by the LPG plant.

Conversion Factor:

Parameter	Unit	Value	Source
Dry gas volume	m^3/mmscf	28,316	HYSYS Software[3]
LPG density	ton/m^3	0.5377	GC measurements
Condensate volume	barrel/ m^3	6.290	HYSYS Software

Result

Parameter	Unit	Total 2nd period	2007	2008
$V_{F,y}$	m^3	75,339,728	60,127,715	15,212,013

[3] HYSYS is powerful software that was created by Hyprotech for simulation of chemical plants and oil refineries. It includes tools for estimation of physical properties and liquid-vapor phase equilibria, heat and material balances, and simulation of many types of chemical engineering equipment.

$NCV_{RG,F,y}$	TJ/m^3	5.0868 x 10^{-05}	5.0868 x 10^{-05}	5.0868 x 10^{-05}
$EF_{CO2,methane}$	tCO_2/TJ	49.55	49.55	49.55
BE_y		189,896	151,554	38,342

4.2 Project Emissions

4.2.1 Formula used for project emissions calculation

The formula used for determination of the baseline emissions are described in section 4.3 of Project Document v.2 dated 5th September 2009.

Project emissions are calculated as follows:

$$PE_y = PE_{CO2,fossilfuels,y} + PE_{CO2,\ elec,y} \qquad (2)$$

Parameter	Description	Unit	Value	Source
PE_y	Project emission in the period y	tCO_2e	Calculated	Equation (2)
$PE_{CO2,fossilfuels,y}$	CO_2 emissions due to consumption of fossil fuels for the recovery, pre-treatment, transportation, and compression of the recovered gas during period y	tCO_2e	Calculated	"Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion"
$PE_{CO2,\ elec,y}$	CO_2 emission due to the use of electricity for recovery, pre-treatment, transportation, and compression of the recovered gas during period y	tCO_2e	0	"Tool to calculate baseline, project and/or leakage emissions from electricity consumption"

Project activity does not use electricity for recovery, pre-treatment, transportation, and compression of the recovered gas. Therefore, $PE_{CO2,\ elec,y} = 0$.

Dry gas (or fuel gas) has been used in the powerhouse to run the gas engines for internal consumption of the gas processing plant. Diesel fuel has also been used to run the diesel engines for emergency situation. Thus, the use of dry gas and diesel fuel are calculated as project emission due to fossil fuel consumption or $PE_{CO2,fossilfuel,y}$. During the entire crediting period $PE_{CO2,fossilfuel,y}$ is calculated according to "Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion" (version 02), where $PE_{CO2,fossilfuel,y} = PE_{FC,j,CO2}$.

$$PE_{FC,j,CO2} = \Sigma FC_{i,j,y} \times COEF_{i,y} \qquad \text{Fossil fuel tool 1)}$$

Parameter	Description	Unit	Value	Source
$PE_{FC,CO2}$	CO_2 emissions from fossil fuel combustion during year y	tCO_2/y	Calculated	Fossil fuel tool 1)
$FC_{i,y}$	Quantity of fuel type i combusted during the year y	Dry gas : m^3/y Diesel fuel: litre	Measured	MLK
$COEF_{i,y}$	CO_2 emission coefficient of fossil fuel type i in year y	Dry gas: tCO_2/m^3 Diesel fuel: tCO_2/ton	Calculated	Fossil fuel tool 4)
i	Fuel types combusted in	-	i=	MLK

	process j during year y		- Dry gas from recovered associated gas (natural gas) - Diesel fuel for diesel gensets	

$COEF_{i,y}$ is calculated using option B. Option B calculates $COEF_{i,y}$ based on net calorific value and CO_2 emission factor of fuel type i, as follows:

$COEF_{i,y} = NCV_{i,y} \times EF_{CO2,i}$ Fossil fuel tool 4)

Parameter	Description	Unit	Value	Source
$COEF_{i,y}$	CO_2 emission coefficient of fossil fuel type i in year y	tCO_2/m^3	Calculated	Fossil fuel tool 4)
$NCV_{i,y}$	Weighted average net calorific value of fuel type i in year		Dry Gas: Measured (GJ/m^3) Diesel Fuel: 0.043 TJ/ton	For dry gas: MLK For diesel fuel: IPCC default
$EF_{CO2,i}$	Weighted average CO_2 emission factor of fuel type i in year y	tCO_2/TJ	Dry Gas: 56.1 Diesel Fuel: 74.1	IPCC 2006 Vol. 2 Chap. 1 Table 1.4
i	Fuel types combusted in process j during year y	-	i= - Dry gas from recovered associated gas (natural gas) - Diesel fuel for diesel gensets	MLK

Project emissions from the consumption of fossil fuels

Using "Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion" (Version 02), the result of $PE_{CO2,fossilfuel,y}$ calculation is tabulated as follows:
$PE_{CO2,fossilfuel,y} = PE_{dry\ gas,CO2} + PE_{diesel\ fuel,CO2}$

$PE_{dry\ gas,CO2}$ calculation

$PE_{dry\ gas,CO2} = FC_{dry\ gas,y} \times COEF_{dry\ gas,y}$
$COEF_{dry\ gas,y} = NCV_{dry\ gas,y} \times EF_{CO2,dry\ gas}$

$PE_{diesel\ fuel,CO2}$ *calculation*

$PE_{diesel\ fuel,CO2} = FC_{diesel\ fuel,y} \times \rho_{diesel\ fuel} \times COEF_{diesel\ fuel,y}$
$COEF_{diesel\ fuel,y} = NCV_{diesel\ fuel,y} \times EF_{CO2,diesel\ fuel}$

4.2.2 Project emissions calculation

Fuel consumption

Date		Month	Year	**Dry Gas (mmscf)**	**Diesel Fuel (litre)**
From	To				
28	31	Mar	2007	8.80	-
1	30	Apr	2007	63.09	10,000
1	31	May	2007	65.03	10,000

1	30	Jun	2007	61.57	10,000
1	31	Jul	2007	63.96	5,000
1	31	Aug	2007	63.57	5,000
1	31	Sep	2007	61.58	-
1	31	Oct	2007	63.60	-
1	30	Nov	2007	59.43	5,000
1	31	Dec	2007	62.77	5,000
1	31	Jan	2008	66.73	5,000
1	28	Feb	2008	52.62	-
1	27	Mar	2008	49.39	10,000
			TOTAL	**742.14**	**65,000**

Conversion Factor:

Parameter	Unit	Value	Source
Dry gas volume	m^3/mmscf	28,316	HYSYS Software
Diesel fuel density	ton/litre	8.15×10^{-04}	Supplier PT. Pertamina

Result

	FC	NCV	EFco2	COEF	PEco2
Fuel type/Unit	*m^3*	*TJ/m^3*	*tCO_2/TJ*	*tCO_2/m^3*	*tCO_2*
2007:Dry Gas	16,236,395	4.2141×10^{-05}	56.10	2.3641×10^{-03}	38,385
2008:Dry Gas	4,778,042	4.2141×10^{-05}	56.10	2.3641×10^{-03}	11,296
Fuel type/Unit	*ton*	*TJ/ton*	*tCO_2/TJ*	*tCO_2/ton*	*tCO_2*
2007:Diesel Fuel	41	4.3000×10^{-02}	74.10	3.1863	130
2008:Diesel Fuel	12	4.3000×10^{-02}	74.10	3.1863	39
			2007: $PE_{CO2,y}$		38,515
			2008: $PE_{CO2,y}$		11,335
			Total $PE_{CO2,y}$		49,850

4.3 Emission Reductions

4.3.1 Formula used for emission reductions calculation

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y \quad (3)$$

Parameter	Description	Unit	Value	Source
ER_y	Project emission in the period y	tCO_2e	Calculated	Equation (3)
BE_y	Baseline emissions in the period *y*	tCO_2e	Calculated	Equation (1)
PE_y	Project emissions in the period *y*	tCO_2e	Calculated	Equation (2)

4.3.2 Emission emissions calculation

	2007	2008	Total	
BE_y	151,554	38,342	189,896	ton CO2
$PE_{CO2,dry\ gas,y}$	38,385	11,296	49,681	ton CO2
$PE_{CO2,diesel\ fuel,y}$	130	39	169	ton CO2

PE_y	38,515	11,335	49,850	ton CO_2
ER_y	113,039	27,007	**140,046**	ton CO_2

5. Environmental Performance

Since the Project commencement, one set of environmental assessment report for the local environmental agency had been developed in May and November 2007, which covers the monitoring period of March 2007 to March 2008. The subsequent environmental performance test was carried out for the monitoring report submission. Table below show the conclusion of the environmental assessment report as summarized in Chapter III of the reports[4].

Parameter monitored	Conclusion
Air Quality	Exhaust gas emission and ambient air qualities on the whole were still fulfilled the environmental quality standard in effect based on South Sumatra Governor Regulation No. 15/2005.
Noise Level	Noise levels measurement that were conducted at every noise sourced and within factory complex in general was still fulfilled the noise level standard for industrial allocation (70±3 dBA).

6. Conclusion

According the management meeting in November 2009, the management of MLK has decided to split the three-year monitoring report into twelve-month reporting. From 28 March 2007 to 27 March 2008, the project activity has performed satisfactorily and achieved a total emission reduction of 140,046 tCO_2, which has been calculated according to the approved methodology set out in the VCS PD version 2. The monitoring of required parameters is in accordance with the approved monitoring methodology as described in the VCS PD.

The company management also believes that its VER Team has taken all reasonable efforts to maintain credibility of data through proper administration and supervision. The project developer therefore wishes to immediately expedite the verification process of the resulting emission reduction.

[4] Please refer to submitted document "UKL UPL 2007 Sem I.doc" and "UKL UPL 2007 Sem II.doc"

Appendix 1 Complete Emission Reduction Xcel Sheet calculation

The complete emission reduction calculation as described in section 4 is following.

Emission Reductions

	2nd period
Start date	28-Mar-07
End date	27-Mar-08

Total	BE_y	189,896	ton CO_2
	$PE_{CO2,dry\ gas,y}$	49,681	ton CO_2
	$PE_{CO2,diesel\ fuel,y}$	169	ton CO_2
	PE_y	49,850	ton CO_2
	ER_y	140,046	ton CO_2
2007	BE_y	151,554	ton CO_2
	$PE_{CO2,dry\ gas,y}$	38,385	ton CO_2
	$PE_{CO2,diesel\ fuel,y}$	130	ton CO_2
	PE_y	38,515	ton CO_2
	ER_y	113,039	ton CO_2
2008	BE_y	38,342	ton CO_2
	$PE_{CO2,dry\ gas,y}$	11,296	ton CO_2
	$PE_{CO2,diesel\ fuel,y}$	39	ton CO_2
	PE_y	11,335	ton CO_2
	ER_y	27,007	ton CO_2

Baseline emission calculation

Baseline Emission

$$BE_y = V_{F,y} \cdot NCV_{RG,F,y} \cdot EF_{CO2,Methane}$$

Parameter	Unit	Total 2nd period	2007	2008
$V_{F,y}$	m3	75,339,728	60,127,715	15,212,013
$NCV_{RG,F,y}$	TJ/m3	5.0868E-05	5.0868E-05	5.0868E-05
$EF_{CO2,methane}$	tCO2/TJ	49.55	49.55	49.55
BE_y		189,896	151,554	38,342

Project Emission calculation

Project Emission from consumption of fossil fuels, $PE_{CO2,fossil\ fuel,y}$

CO2 emissions due to comsumption of fossil fiels, including the associated gas if applicable, for the collection, transportation and processing of the associated gas

$$PE_{FC,j,y} = \sum_i FC_{i,j,y} \times COEF_{i,y} \quad (1)$$

$$COEF_{i,y} = NCV_{i,y} \times EF_{CO2,i,y} \quad (4)$$

Type of fossil fuels consumed in the project activity:

1. Dry gas (or fuel gas), which has been used in the powerhouse to run the gas engines for internal consumption of the gas processing plant.
2. Diesel fuel, which has also been used to run the diesel engines for emergency situation.

	FC	NCV	EF_{CO2}	COEF	PE_{CO2}
Fuel type/Unit	*m3*	*TJ/m3*	*tCO2/TJ*	*tCO2/m3*	*tCO2*
2007:Dry Gas	16,236,395	4.2141E-05	56.10	2.3641E-03	38,385
2008:Dry Gas	4,778,042	4.2141E-05	56.10	2.3641E-03	11,296
Fuel type/Unit	*ton*	*TJ/ton*	*tCO2/TJ*	*tCO2/ton*	*tCO2*
2007:Diesel Fuel	41	4.3000E-02	74.10	3.1863E+00	130
2008:Diesel Fuel	12	4.3000E-02	74.10	3.1863E+00	39
			2007: $PE_{CO2,y}$		38,515
			2008: $PE_{CO2,y}$		11,335
			Total $PE_{CO2,y}$		49,850

Input Parameters:

Volume of total recovered gas, $V_{F,y}$

Parameter		Date: From	Date: To	Month	Year	$V_{feed\ gas\ supplied,y}$ (mmscf)	$V_{dry\ gas,y}$: Delivered back to MEPI (mmscf)*	$V_{dry\ gas,y}$: Total Fuel-own cons. (mmscf)**	$V_{LPG,y}$ ton	$V_{condensate,y}$ barrel
Volume of total feed gas supplied	$V_{feed\ gas\ supplied,y}$	28	31	Mar	2007	57.84	32.93	8.80	356.24	1,547.84
		1	30	Apr	2007	378.09	206.63	63.09	2,438.04	9,081.78
		1	31	May	2007	335.87	176.45	65.03	2,195.80	7,855.16
		1	30	Jun	2007	330.30	168.73	61.57	2,125.93	7,456.31
		1	31	Jul	2007	324.26	168.30	63.96	2,086.66	7,094.43
		1	31	Aug	2007	321.69	159.90	63.57	2,106.79	7,171.81
		1	31	Sep	2007	312.38	156.43	61.58	2,025.75	6,852.67
		1	30	Oct	2007	306.60	150.50	63.60	1,974.29	6,812.41
		1	31	Nov	2007	307.97	166.41	59.43	1,746.35	6,323.26
		1	30	Dec	2007	300.14	162.17	62.77	1,730.53	5,661.13
		1	31	Jan	2008	289.06	141.57	66.73	1,657.98	5,897.87
		1	31	Feb	2008	214.51	119.92	52.62	1,300.31	4,812.74
		1	27	Mar	2008	188.28	106.64	49.39	1,127.06	4,399.09
				2007	**TOTAL**	2,975.14	1,548.45	573.40	18,786.38	65,856.80
				m3	**TOTAL**	84,244,065	43,845,911	16,236,395	34,938	10,471
				2008	**TOTAL**	691.85	368.13	168.74	4,085.35	15,109.70
				m3	**TOTAL**	19,590,510	10,423,970	4,778,042	7,598	2,403

* Dry gas delivered to MEPI is also called lean gas.

** Dry gas used as fuel for own consumption is also called fuel gas.

2007: Volume of total feed gas supplied	$V_{feed\ gas\ supplied,y}$	m3	**TOTAL**	84,244,065	
2008: Volume of total feed gas supplied	$V_{feed\ gas\ supplied,y}$	m3	**TOTAL**	19,590,510	103,834,575
2007: Volume of flared gas	$V_{flare,y}$	m3	**TOTAL**	24,116,350	
2008: Volume of flared gas	$V_{flare,y}$	m3	**TOTAL**	4,378,497	28,494,847
2007: Volume of total recovered gas	$V_{F,y}$	m3	**TOTAL**	60,127,715	
2008: Volume of total recovered gas	$V_{F,y}$	m3	**TOTAL**	15,212,013	75,339,728

Conversion Parameter

Dry Gas	1 mmscf =	28,316	m3	Source: HYSYS Software
	btu/scf =	3.7258E-08	TJ/m3	
LPG	density =	0.5377	t/m3	Source: Gas Chromatography measurement, please refer to sheet "LPG density"
Condensate	1 m3 =	6.290	barrel	Source: HYSYS Software

Characterisation of recovered gas

NCV of recovered gas, calculated from feed gas composition	$NCV_{RG,F,y}$	btu/scf	**Average**	1365.30
Please refer to sheet of "Feed Gas Composition & NCV"		TJ/m3	**Average**	5.0868E-05
CO_2 emission factor for methane	$EF_{CO2,methane}$	tCO_2/TJ	49.55	

Feed Gas Composition & NCV

Credit. period	Month	btu/scf
2nd	Apr 07-Mar 08	1365.30

Month/Year	2007	2008
Jan		1103.47
Feb		1388.71
Mar	1393.53	1362.99
Apr	1397.68	
May	1384.32	
Jun	1390.18	
Jul	1418.99	
Aug	1389.64	
Sep	1384.58	
Oct	1401.71	
Nov	1343.07	
Dec	1390.01	

LPG relative density

Credit. period	Month	RD	Density
2nd	Apr 07-Mar 08	0.5377	537.7

Month/Year	Date	2007	Date	2008
Jan			17	0.5416
Feb			17	0.5389
Mar	24	0.5408	17	0.5384
Apr	17	0.5362		
May	17	0.5362		
Jun	17	0.5372		
Jul	17	0.5361		
Aug	17	0.5341		
Sep	17	0.5354		
Oct	17	0.5387		
Nov	17	0.5388		
Dec	17	0.5379		

Consumption of Fossil Fuel

Type of fuels	Consumption	Date		Month	Year	Dry Gas (mmscf)
		From	To			
Dry Gas		28	31	Mar	2007	8.80
		1	30	Apr	2007	63.09
		1	31	May	2007	65.03
		1	30	Jun	2007	61.57
		1	31	Jul	2007	63.96
		1	31	Aug	2007	63.57
		1	31	Sep	2007	61.58
		1	30	Oct	2007	63.60
		1	31	Nov	2007	59.43
		1	30	Dec	2007	62.77
		1	31	Jan	2008	66.73
		1	31	Feb	2008	52.62
		1	27	Mar	2008	49.39
					Total	**742.14**
	2007: Dry Gas consumption			FCdry gas,y	m3	16,236,395.00
	2008: Dry Gas consumption			FCdry gas,y	m3	4,778,042.00
	NCV of dry gas, average			NCVdrygas,y	TJ/m3	4.2141E-05
	Please refer to sheet of "Dry Gas Composition & NCV"				btu/ft3	1131.09
	CO2 emission factor of fuel type i			$EF_{CO2,dry\ gas}$	tCO_2/TJ	56.1

Type of fuels	Consumption	Date		Month	Year	Diesel Fuel (litre)
		From	To			
Diesel Fuel	Diesel Fuel	28	31	Mar	2007	-
		1	30	Apr	2007	10,000
		1	31	May	2007	10,000
		1	30	Jun	2007	10,000
		1	31	Jul	2007	5,000
		1	31	Aug	2007	5,000
		1	31	Sep	2007	-
		1	30	Oct	2007	-
		1	31	Nov	2007	5,000
		1	30	Dec	2007	5,000
		1	31	Jan	2008	5,000
		1	31	Feb	2008	-
		1	27	Mar	2008	10,000
					TOTAL	**65,000**
	2007:Diesel Fuel consumption			FCi,y	ton	41
	2008:Diesel Fuel consumption			FCi,y	ton	12
	NCV of diesel fuel, IPCC default				TJ/ton	0.043
	Density of diesel fuel, supplier (Pertamina)				kg/litre	0.815
	CO2 emission factor of fuel type i			$EF_{CO2,diesel\ fuel}$	tCO_2/TJ	74.1

Dry Gas NCV

Credit. period	Month	btu/scf
2nd	Apr 07-Mar 08	1131.09

Month/Year	2007	2008
Jan		1131.10
Feb		1217.72
Mar	1121.52	1118.58
Apr	1119.10	
May	1122.86	
Jun	1124.74	
Jul	1125.27	
Aug	1127.34	
Sep	1127.67	
Oct	1123.16	
Nov	1114.01	
Dec	1131.17	